

# **Web Development Assignment – 2**

**Topic:**

**React.js, Components, State Management & Navigation**

**Weather Forecast System:**

**Student Details**

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**Technology:** React.js

**Assignment:** React Frontend Development

## **1. Objective**

The objective of this assignment is to develop a responsive frontend interface for the Weather Forecast System using React.js. The application is designed to provide users with real-time weather information and a 5-day weather forecast by integrating with the WeatherAPI. The project also demonstrates the use of React components, state management, and routing.

## **2. Technologies Used**

- React.js
- React Router DOM
- JavaScript (ES6)
- HTML5
- CSS3
- WeatherAPI
- Vite
- Visual Studio Code

## **3. Project Overview**

The Weather Forecast System is a React-based web application that allows users to search for any city and view its current weather conditions along with a 5-day weather forecast. The application is designed with a clean, modern, and responsive user interface to provide a good user experience on both desktop and mobile devices.

## **4. React Components Used**

The application is divided into reusable React components for better code organization and maintainability.

- Navbar Component
- SearchBar Component
- CurrentWeather Component
- Forecast Component
- ForecastCard Component
- Footer Component

- Home Page
- About Page

## 5. State Management

React Hooks (useState) are used for managing application state.

The following states are maintained:

- City Name
- Current Weather Data
- 5-Day Forecast Data
- Loading Status

State updates automatically whenever the user searches for a different city.

## 6. Navigation

React Router is used for client-side navigation.

The application contains two pages:

- **Home** – Displays the weather search interface and forecast.
- **About** – Displays project information, features, and technologies used.

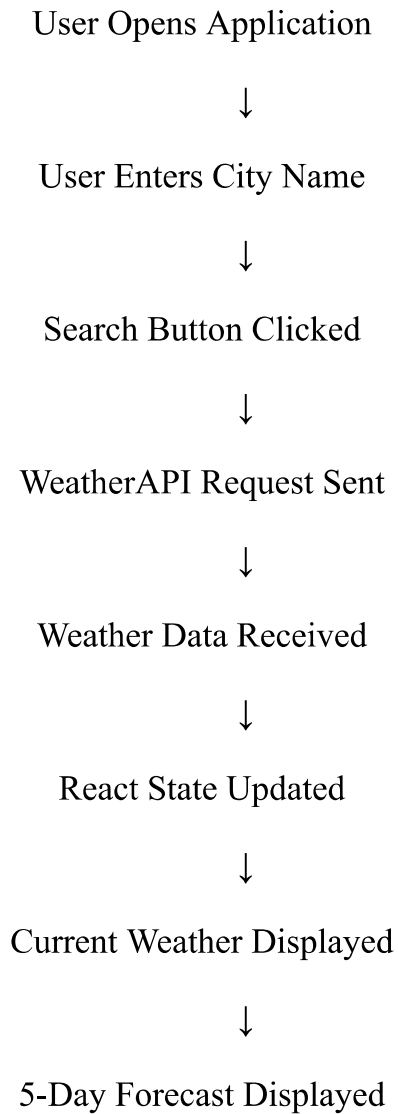
Navigation is handled without refreshing the webpage.

## 7. Features Implemented

- Responsive User Interface
- Search Weather by City Name
- Display Current Temperature
- Weather Condition
- Humidity
- Wind Speed
- Atmospheric Pressure
- 5-Day Weather Forecast
- Modern Glassmorphism UI
- React Router Navigation

- Responsive Design

## 8. Project Workflow



## **9. Observations**

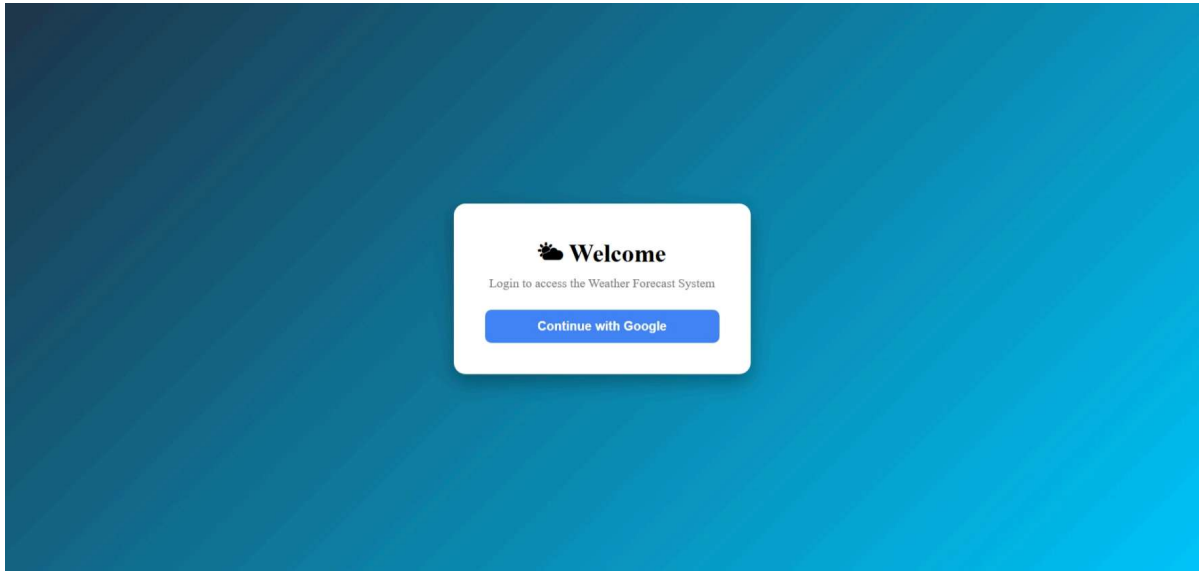
During this assignment, I learned how to build a frontend application using React.js by creating reusable components and managing data with React Hooks. I also learned how to implement navigation using React Router and integrate a third-party API to display real-time weather information. Additionally, I gained practical experience in designing a responsive and user-friendly interface using HTML, CSS, and JavaScript.

## **10. Conclusion**

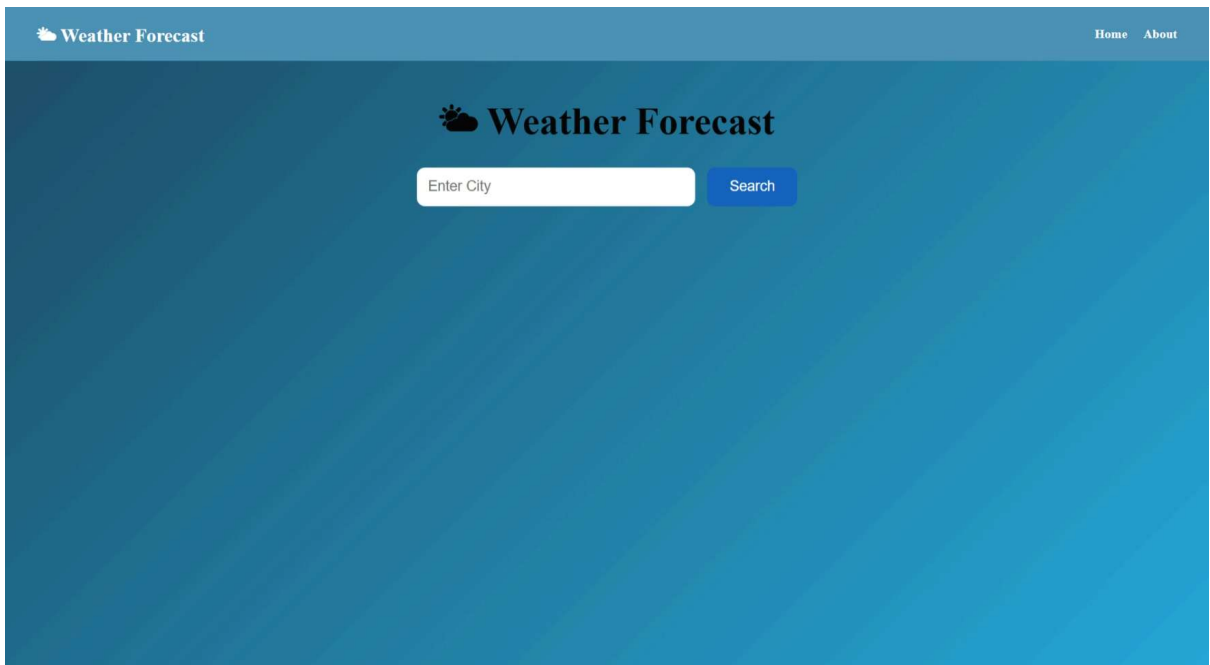
The Weather Forecast System was successfully developed using React.js. The application provides current weather information and a 5-day weather forecast through a responsive and interactive user interface. This assignment improved my understanding of React components, state management, routing, API integration, and modern frontend development practices.

## 11. Screenshots of the UI

### Login Page:



### Home page:



The image shows a web-based weather forecast application. At the top, a dark blue header contains a cloud icon and the text "Weather Forecast" on the left, and "Home" and "About" links on the right. Below the header, the main content area has a lighter blue background. It starts with a "Weather Forecast" title and a cloud icon. A search bar with the text "surat" and a blue "Search" button is positioned below. The next section, "Current Weather", features a globe icon and a white card displaying a sun-behind-a-cloud-with-rain icon, a large temperature of 29.5°C, and details like Humidity (77%), Wind (33.1 km/h), Pressure (1004 mb), and Visibility (10 km). Following this is the "5-Day Forecast" section with a calendar icon, showing five days (July 11-15) with icons, temperatures, and humidity levels.

About page:

